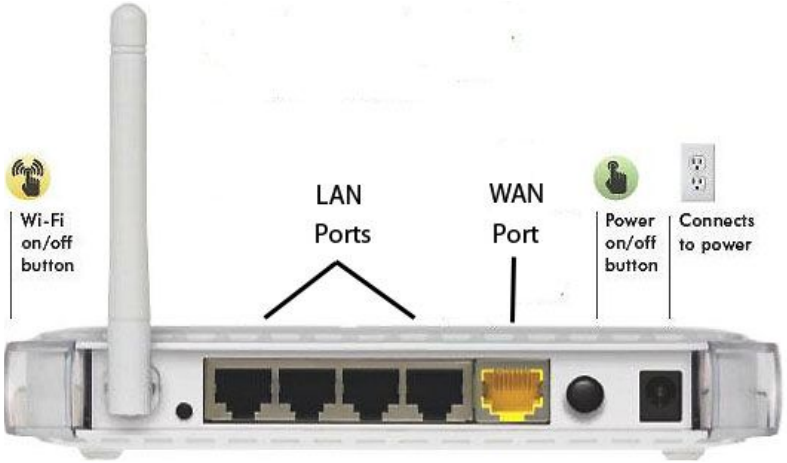
Introduction to Router

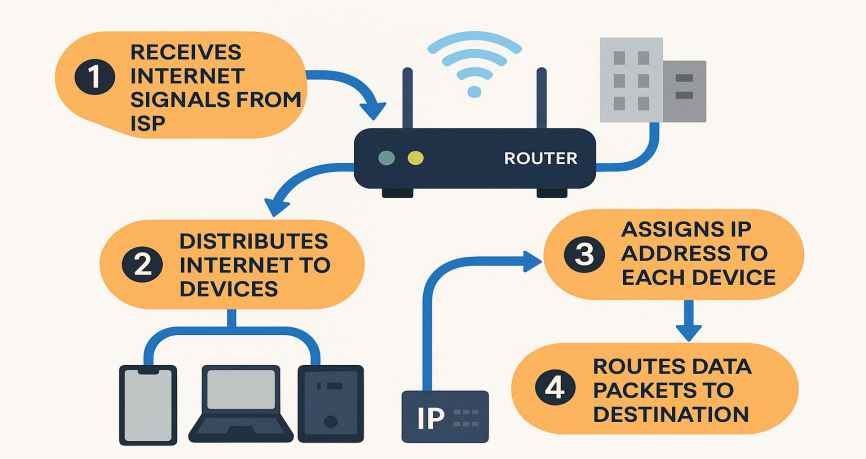
A router is a networking device that forwards data packets between computer networks. It connects multiple networks, such as your home network to the internet, or different parts of a large business network. Routers work by examining the destination address of each data packet and then directing it to the appropriate network or device.



How does a router works

How Does a Router Work?

- The **router receives internet signals** from the ISP (Internet Service Provider).
- It then **distributes the internet** to multiple devices using **Wi-Fi or LAN cables**.
- The router **identifies each device** by assigning it a unique IP address.
- It **routes data packets** by checking where they're coming from and where they're going, ensuring they reach the correct destination.
- NAT- Converts private ip into Public ip



Main Functions of a Router

Internet Sharing: Distribute Internet from one line to multiple devices.

Data Routing: Determines where data comes from and where it should go

IP Address Assignment: Assigns a unique IP address to each connected device

Security Provision: Provides protection using firewalls and passwords.

Wi-Fi Network Creation: Enables wireless internet access without physical cables

CISCO ROUTING CONFIG MODE:

- EXE Mode
- Privilege Mode
- Global configuration Mode

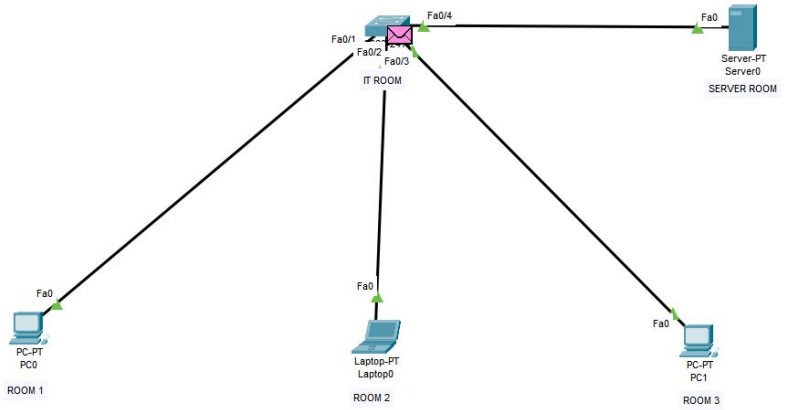
❑ **EXE Mode:** The user EXEC mode of Cisco routers is the normal operating mode. After the Cisco device boots up, the IOS (Internetwork Operating System) loads and enters EXEC mode. The symbol for EXEC mode is “>”. We can check version there.

❑ **Privilege Mode:** The advanced operating mode of Cisco routers is the privileged mode. The symbol for privileged mode is '#'. In this mode, all types of router configurations can be viewed. Router configurations can be copied, saved, or the router can be reloaded from this mode

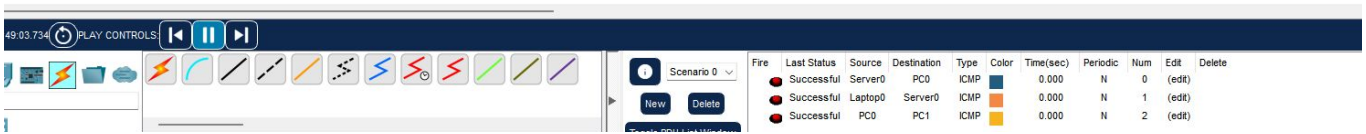
❑ **Global Configuration Mode:** Global configuration mode is the operation mode where any configuration command given affects the entire device. However, to enter global configuration mode, you first need to enter privileged mode. All tasks related to the router are performed in this mode.

DHCP- (Dynamic Host Configuration Protocol)

DHCP is a network protocol that automatically assigns IP addresses and other network settings (like subnet mask, gateway, DNS) to devices when they connect to a network. It uses a client-server model, where the DHCP server manages a pool of IPs and leases them to clients temporarily.



DHCP WITH SERVER



Routing

Routing: Routing is an important part of networking, which primarily determines the optimal path for data packets to travel from one network to another, or from one device to another.

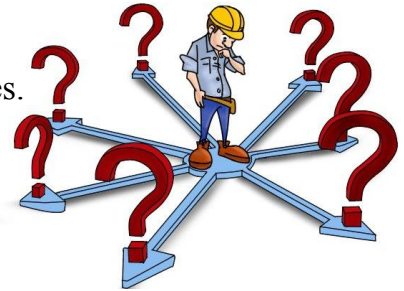


Steps of the Routing Process:

- **Route Discovery** – Determining which path the data will take.
- **Route Selection** – Choosing the most suitable path if multiple routes are available.
- **Packet Forwarding** – Sending the data along the selected path.

Types of Routing:

- **Static Routing** – Routes are configured manually. Typically used in small networks.
- **Dynamic Routing** – Routers automatically learn routes. Commonly used in large networks.
- **Default Routing** – A default path is provided for unknown networks. Usually applied in offices.



Routing Types:

Static Routing

Static Routing is a process where we manually tell the router "To reach this network, go through this specific path." That means each route must be configured by hand. No protocol automatically learns or updates IP addresses everything is done manually. This method is very effective in small networks, where there are fewer routes and everything is manageable manually. It’s also very secure, because no one from outside can change the routes.However, if a link or router fails, you have to manually update the route it doesn’t update automatically.

Dynamic Routing

Dynamic Routing is the opposite of static. Here, routers automatically share their routing information with each other and determine the best path on their own. They use different routing protocols like RIP, OSPF, EIGRP, BGP, etc., to do this. This type of routing is used in large networks, where there are hundreds or thousands of routers. Each router informs its neighbors about its IP addresses and status. If a path fails, the routers automatically find an alternate route. However, dynamic routing requires more CPU and RAM, and the configuration is a bit more complex.

Advantages of Static Routing:

- **Simple & Secure** — When configured manually, no one else can modify it.
- **No Overhead** — Doesn’t consume CPU or bandwidth resources.
- **Greater Control** — You decide exactly which path the data will take.

Disadvantages:

- **Difficult to maintain** in large networks.
- If a **router link goes down**, it doesn’t self-heal—manual updates are required.

Advantages of Dynamic Routing:

- **Self-healing** — If any link fails, it automatically finds a new path.
- **Best for Large Networks** — Even with thousands of connected routers, it remains manageable.
- **Automatic Updates** — Routing tables update themselves without manual intervention.

Disadvantages:

- **Consumes CPU, RAM & Bandwidth** — Because routers constantly share routing information with each other.
- **Configuration is Complex** — Can be tricky for beginners to set up and manage.

Dynamic Routing Protocols:

In dynamic routing, protocols are used to exchange route information between routers.

There are two types:

- Distance Vector Protocols Example: RIP (Routing Information Protocol) Suitable for slower and smaller networks.Selects routes based on hop count.
- Link State Protocols Example: OSPF (Open Shortest Path First) Chooses routes based on network topology. Performs well in faster and larger networks.

Routing Metrics

- **Hop Count**: Indicates how many routers (hops) a packet has to pass through to reach its destination.
- **Bandwidth**: Refers to the data transfer rate of the path—the higher the bandwidth, the better the path.
- **Delay**: Measures the time it takes for data to travel along a route—less delay means a faster path.
- **Cost:** A value assigned by the routing protocol that reflects preference; it can be set manually or calculated automatically

Types of Routing Protocol

1. RIP
2. OSPF
3. EIGRP

1. RIP (Routing Information Protocol): RIP is an old and simple routing protocol that selects routes based on **hop count**

How it works:

- Each router says: “I know this network. To reach it through me, you’ll need to pass through 2 routers.”
- The route with the **lowest hop count** is chosen.
- No need to be an expert—RIP has a **maximum hop limit of 15**. If the route requires more than 15 hops, it’s considered **unreachable**.

Example: Suppose your PC needs to reach a server. One path goes through 3 routers, another through 2. RIP will say: “Let’s go through 2 hops — that’s the shortest!”

Problems:

- **Slow and inefficient** in large networks
- Can cause **routing loops**
- **Delayed updates** may lead to outdated routing info

2. OSPF (Open Shortest Path First)

OSPF is a **modern, fast, and intelligent routing protocol** that selects the best route between routers based on **link cost or metrics**.

How it works:

- Routers create a **map** called a *Topology Table*.
- Then, using the **Dijkstra Algorithm**, they calculate which path is the most efficient or least costly.
- It's a **Link-State Protocol**, meaning routers inform each other: "These are the links I have and here's their status."

Example: Suppose there are three routes, one with 10 Mbps and another with 1 Gbps speed — OSPF will choose the route with **higher speed and lower delay**.

Advantages:

- Highly **efficient in large networks**
- **Fast convergence** — quickly learns new routes
- **Prevents routing loops**

3. EIGRP (Enhanced Interior Gateway Routing Protocol)

EIGRP is **Cisco's proprietary routing protocol**. It's a **hybrid protocol**, combining features of both **Distance Vector** and **Link-State** routing.

How it works:

- It evaluates multiple factors like **delay, bandwidth, load, and reliability** to determine the best route.
- Makes decisions **quickly and intelligently**.

Example: If you're trying to reach Google, EIGRP will say: "This route has higher bandwidth and lower delay — take this one."

Advantages:

- **Fast convergence** — quickly adapts to network changes
- **Reliable** — ensures stable routing decisions
- Works **only on Cisco routers**